Irvin Lim

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About Me

Staff-level distributed systems engineer, specializing in Kubernetes, workload scheduling and Linux resource isolation.

Tech Lead of the **Batch Colocation Platform** at Shopee, which runs batch workloads safely alongside production services on opportunistic idle capacity. Today the platform runs hundreds of thousands of batch workloads on 1M+ reclaimed CPU cores and PBs of memory across more than 40,000 nodes fleet-wide.

Professional Experience

Shopee — Singapore

Senior Expert Engineer, Unified Scheduling — *Jul 2025 – Present*

- As technical lead of the **Batch Colocation Platform**, delivered tens of millions of dollars in estimated annual savings across six major customers within Big Data, Content and Marketplace businesses, onboarding new tenants and driving cross-team design changes, which raised reclaimed CPU 4x and memory 6x within two years.
- Owned and led a cross-org **Reverse Colocation** program to provide burst capacity of 300K+ CPUs for *Marketplace* during sales campaigns, designing migration mechanisms to preempt lower-priority Big Data workloads. Scaled coverage to 80% across 4 major campaigns within half a year, avoiding the need to purchase thousands of additional machines.
- Led the evolution of **Eunomia Agent**, a node agent deployed fleet-wide across tens of thousands of nodes, extending Kubelet behaviour and implementing specialized cgroups, cpusets, NUMA and resctrl allocation policies, providing customized isolation capabilities across shared and dedicated latency-sensitive pools.
- Drove high-profile incident response efforts, including resolving long-standing system-wide stalls due to page thrashing from a mix of cpuset/NUMA placement, `mpol` policy, and `memcg` watermarks; led efforts to investigate and remediate a Go memory corruption bug where FD misuse caused BPF syscalls to mangle the userspace heap.
- Drove the architecture and adoption of **Global Scheduler**, a cross-cluster scheduler that manages 100,000+ pods across 50+ services on 40+ clusters. Led critical fixes for head-of-line blocking and flip-flops, achieved 100% rollout coverage for the *Content* business and unlocked cross-cluster autoscaling to cut off-peak resource waste by up to 3x.
- Established roadmaps to drive continued resource cost and placement optimization projects across the organization. Mentored 10+ teammates through weekly 1:1s, and oversaw the growth of four ICs into component owners.

Expert Engineer, Unified Scheduling — *Jan 2023 – Jun 2025*

- Led resource delivery efforts for **Livestream Colocation**, designing and validating isolation mechanisms for transcoding workers colocated alongside production services to support the growth of *Shopee Live*. Drove aggressive overcommit and migration efforts to move off costly public cloud and grow transcoding capacity 8x within a quarter.
- Designed PSI-based eviction and rescheduling strategies to allow lower-priority live transcoding workloads to safely run on opportunistic batch capacity, validated through eviction tuning, load tests and greyscale rollouts. Scaled batch transcoding capacity to 400K CPU cores in half a year without needing additional increases in resource budgets.
- Drove the isolation evaluation for **Dedicated Pool Colocation** with latency-sensitive *Search, Recommendation & Ads* pools, proposing SMT-aware isolation over static core reservation to contain noisy neighbor interference, and led the stress testing and phased rollout that raised CPU utilization from 15% to 60% with <1 ms added P99 latency.
- Designed and implemented a colocation management plane that centralized configuration scattered across node agents, schedulers and controllers into policy-driven delivery, and standardized releases on 100% of clusters via ArgoCD within a quarter, replacing error-prone manual configuration and eliminating configuration drift across hundreds of clusters.

Senior Engineer, Unified Scheduling — *Jan 2022 – Dec 2022*

- Developed core colocation functionality in **Eunomia Agent**, including a dual-scheduler YARN–K8S model to minimize adoption costs, and unified two legacy cgroup agents into one, unblocking colocation rollout.

- Drove the initial adoption of **YARN Colocation**, running low-priority Apache Spark and Flink jobs alongside production traffic with CPU, memory, I/O and memory-bandwidth isolation (cgroups, resctrl) that raised peak CPU utilization fourfold with <5% P99 impact, providing estimated cost savings of over a million dollars in its first year.

Software Engineer, Linux Elastic Application Platform *May 2020 – Dec 2021*

- Designed and built **Runonce**, a Kubernetes-native, exactly-once cron and ad-hoc job scheduler that powers millions of job executions per day in production, replacing a legacy scheduler prone to crashing every 2–4 weeks under load.
- Migrated tens of thousands of jobs across the entire company to the new scheduler within a single quarter, while simultaneously supporting 3x year-on-year growth in job volume, and cutting tail task launch latency from 2 minutes to 10 seconds via jittered scheduling techniques.

Internships

Facebook — Menlo Park, CA, USA

Production Engineer Intern, Data Infrastructure *May 2019 – Aug 2019*

- Developed a web service for managing hundreds of production PrestoDB clusters, which replaced the old Presto dashboard with 15x faster load time.

Shopee — Singapore

Software Engineering Intern, TechOps *May 2018 – Oct 2018*

- Designed and developed the first version of **Bromo**, a Mesos-based container orchestrator that evolved onto Kubernetes and now runs millions of pods company-wide.

Conference Talks

Squeeze Your K8s: How We Adopt Time-Series Forecasting Models in FinOps Practices
KubeCon + CloudNativeCon Europe 2024, Paris, France

Colocating Hadoop YARN with Kubernetes to Save Massive Costs on Big Data
KubeCon + CloudNativeCon Europe 2023, Amsterdam, Netherlands

Education

Brown University — Providence, RI, USA

Master of Science, Computer Science *Jan 2019 – May 2020*

National University of Singapore — Singapore

Bachelor of Computing (Honours), Computer Science *Aug 2015 – Dec 2018*

Skills

- **Languages**: Go (primary), C, Python, Java, JavaScript
- **Linux Kernel**: Performance profiling (eBPF, perf, PSI, RDT), runtime isolation (cgroups, resctrl, NUMA/SMT), Linux task scheduling, Linux memory management
- **Resource Management**: FinOps, workload colocation, multi-tenancy and QoS, overcommit, eviction and preemption, capacity planning, time-series forecasting
- **Kubernetes**: Cluster management, application delivery (Helm, ArgoCD), customized controller and scheduler plugin development, node & container runtime (Kubelet, Topology Manager, CPU Manager, Docker, containerd, NRI)
- **Distributed Systems**: Multi-level and job scheduling, cluster federation, high availability and leader election, service discovery and load balancing
- **Observability & Data**: Prometheus, VictoriaMetrics, Grafana, MySQL, PostgreSQL, Redis
- **Language Fluency**: English (native), Mandarin Chinese (native)